

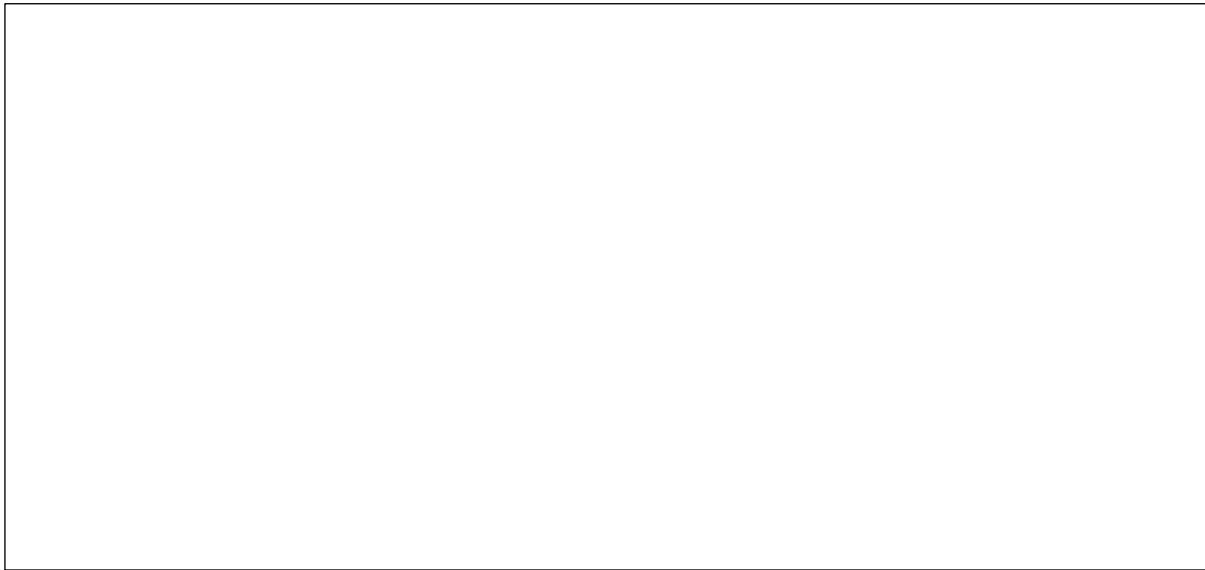
Lab 08 – Worksheet

Programming Basys-3 Board

Name:	ID:	Section:
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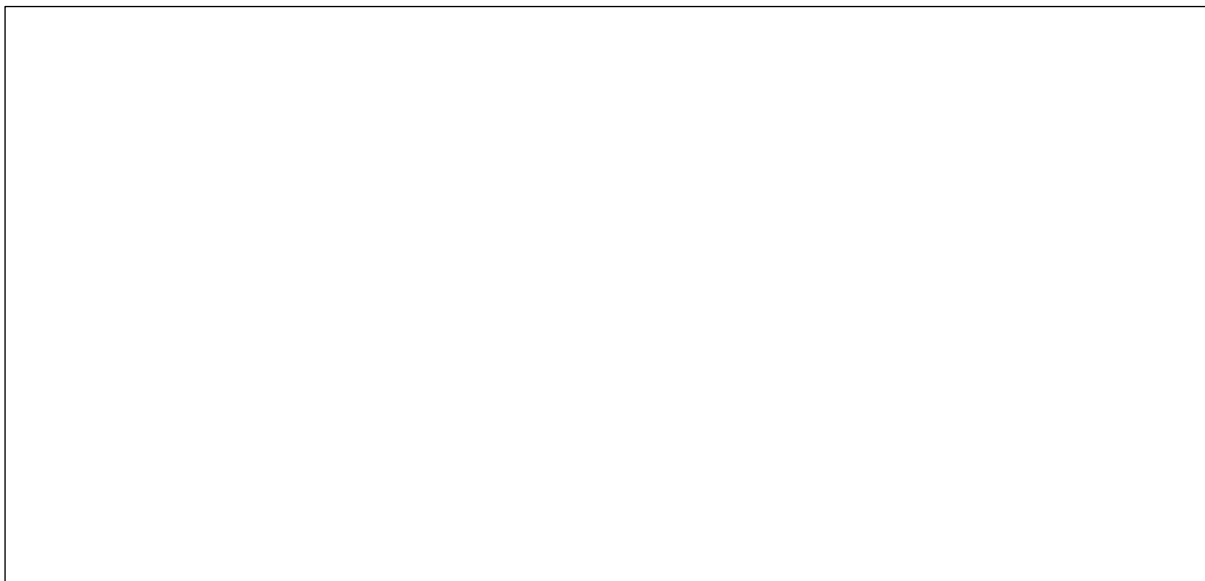
Task a) Creating Design Modules & Testbench on Vivado Design Suite

Attach screenshot of timing diagram of AND gate.

**Task b) Programming Basys-3 FPGA Board**

Attach snapshot of simulation output of AND gate (Picture of your FPGA Board).

Note: The picture must not be blurry.



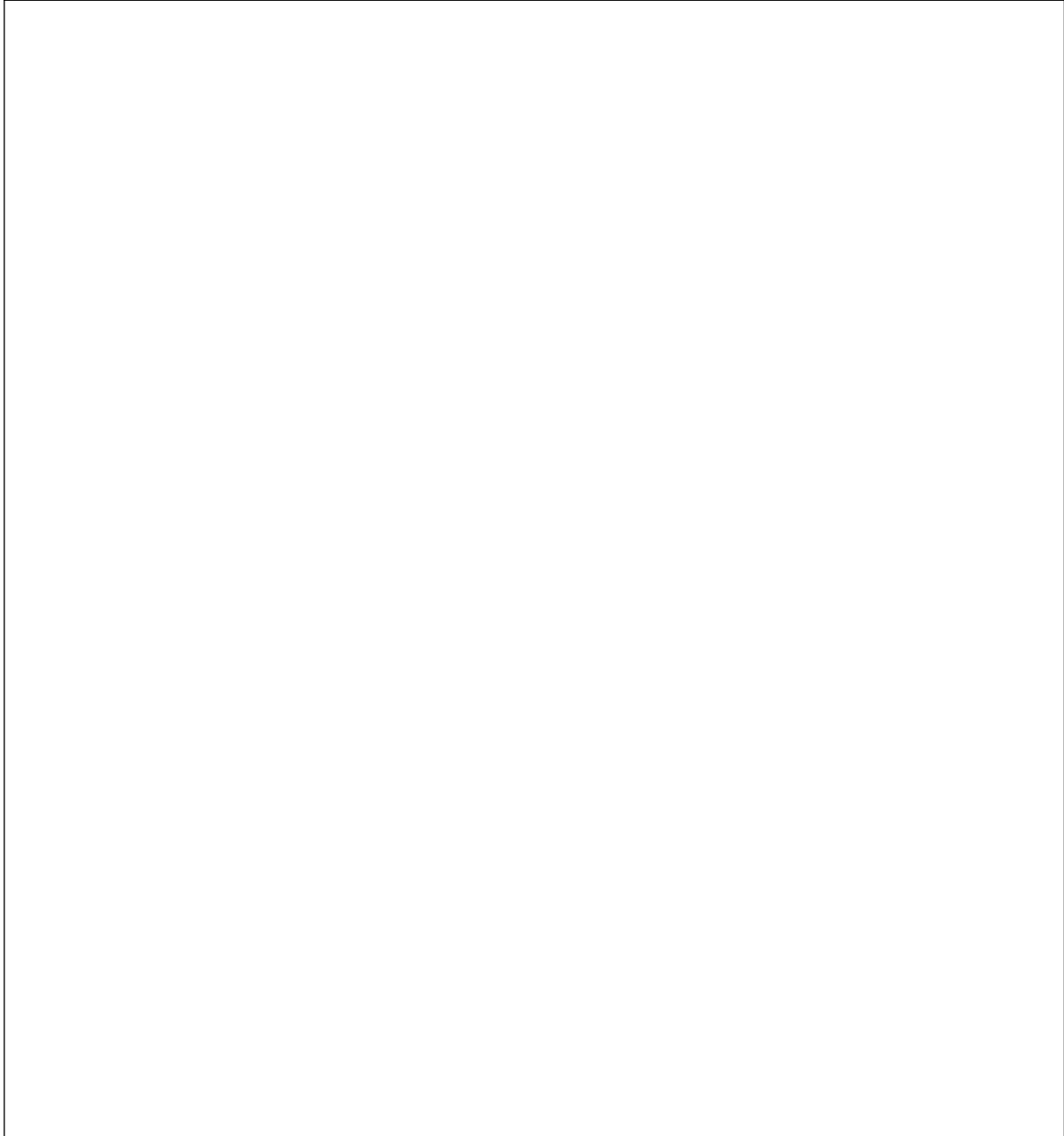
Task c) Testing Adder/Subtractor Design on Basys-3 FPGA Board

- (i) Provide appropriately commented code of **all** design module of Adder/Subtractor Design (Testbenches are not required).

(ii) Provide Constraint Files.

- (iii) Attach picture of your programmed FPGA Board with output displayed on 7 – segment display and write the details of output cases displayed. Add at least 3 cases, i.e., addition of inputs, subtraction of inputs with a positive result, and subtraction of inputs with a negative result.

Note: The picture must not be blurry.



Assessment Rubric Lab 08 - Worksheet Programming Basys-3 Board

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Marks Distribution:

		LR2 Code	LR5 Results	LR9 Report
In - Lab	Task a	-	10	10
	Task b	-	10	
	Task c	20	10	30
Total Marks: 100		/20	/30	/40
CLO Mapping		CLO3	CLO3	CLO4

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission and Viva	/10	CLO 4

CLO	Total Points	Points Obtained
3	50	
4	50	
Total	100	

For description of different levels of the mapped rubrics, please refer to the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Assessment Rubrics

#	Assessment Elements	Level 1: Unsatisfactory	Level 2: Developing	Level 3: Good	Level 4: Exemplary
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation/network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR9	Report	All the in-lab tasks are not included in report.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included.	Good Summary of most of the in-lab tasks is included in report. The work is supported by figures and plots with explanation.	Detailed Summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots, and tables.
AR7	Report Submission & Viva	Most of the questions are not answered/ figures are not labelled/titles are not mentioned/ units are not mentioned. No comments are present in the code.	Some of the questions are answered, figures are labelled, titles are mentioned and units are mentioned. Few comments are stated in the code.	Majority of the questions are answered, figures are labelled, titles are mentioned and units are mentioned. Comments are stated in the code.	All the questions are answered, figures are labelled, titles are mentioned and units are mentioned. Proper comments are stated in the code.